

HR Employee Performance and Attrition Analysis

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Introduction

- HR Employee Performance and Attrition Analysis focuses on studying employee behavior, performance, and workplace trends within an organization
- It helps organizations understand factors affecting employee retention, productivity, and job satisfaction
- The analysis includes important HR factors such as salary, promotions, overtime, and work-life balance
- HR analytics supports better workforce planning and employee management through data-driven insights

Problem Statement

- High employee attrition and inconsistent performance management are critical challenges faced by modern organizations.
- Without proper analysis of HR data, companies are unable to identify why employees leave, who is underperforming, and whether compensation is fair, leading to poor retention, higher hiring costs, and reduced productivity.



Objectives

- To understand employee attrition patterns and the factors influencing workforce turnover
- To evaluate employee performance, satisfaction, and work-life balance across different departments
- To analyze salary distribution, promotion trends, and compensation growth within the organization



Solution

- Built a 2-page HR analytics dashboard using Python and Power BI
- Preprocessed HR data using Python
- Analyzed attrition, performance, salary, and promotion trends
- Added interactive charts, KPI cards, and slicers
- Helped identify high-risk employees and workforce patterns

Data Preprocessing



- Removed unnecessary columns and duplicate records
- Handled missing values and cleaned text inconsistencies
- Standardized categorical values for accurate analysis
- Corrected logical inconsistencies in employee records
- Created analytical features like Age Groups, Salary Bands, Promotion Gap, and Attrition Flag

Dashboards



HR EMPLOYEE PERFORMANCE AND ATTRITION ANALYSIS

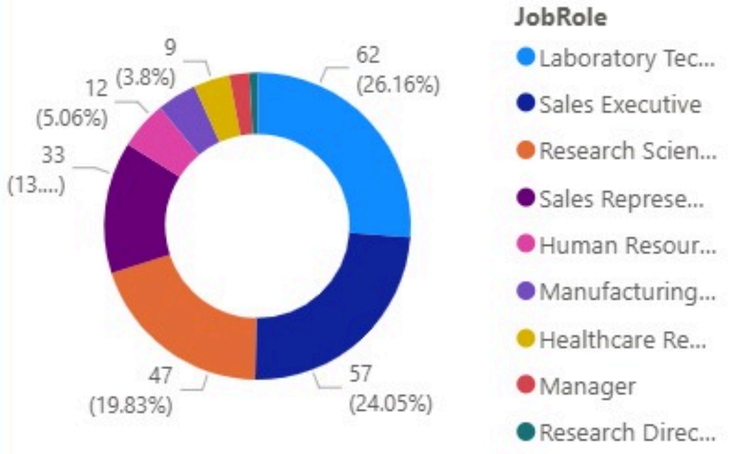
Department
- Sales
- Research & Development
- Human Resources

Gender
- ☐ Female
- ☐ Male

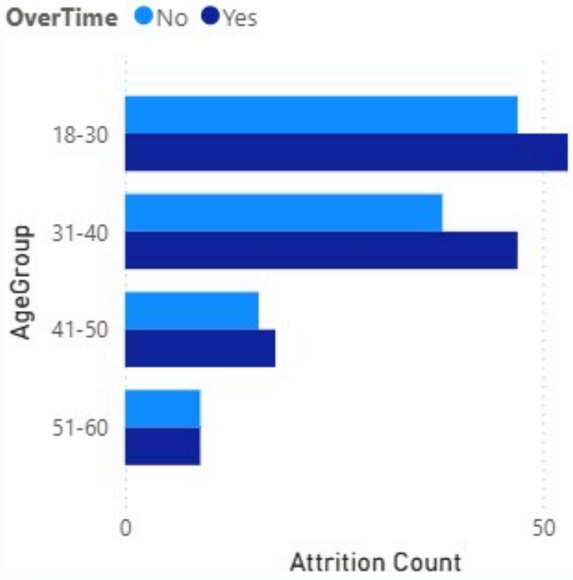
Attrition and Performance

1K	237	0.16	3.15	2.73
Total Employees	Attrition Count	Attrition Rate	Average of PerformanceRa...	Average of JobInvolvement

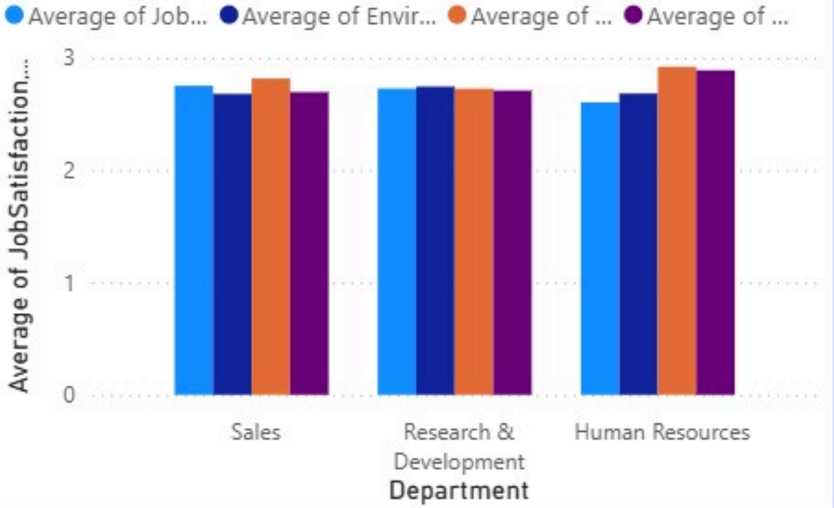
Attrition Count by JobRole



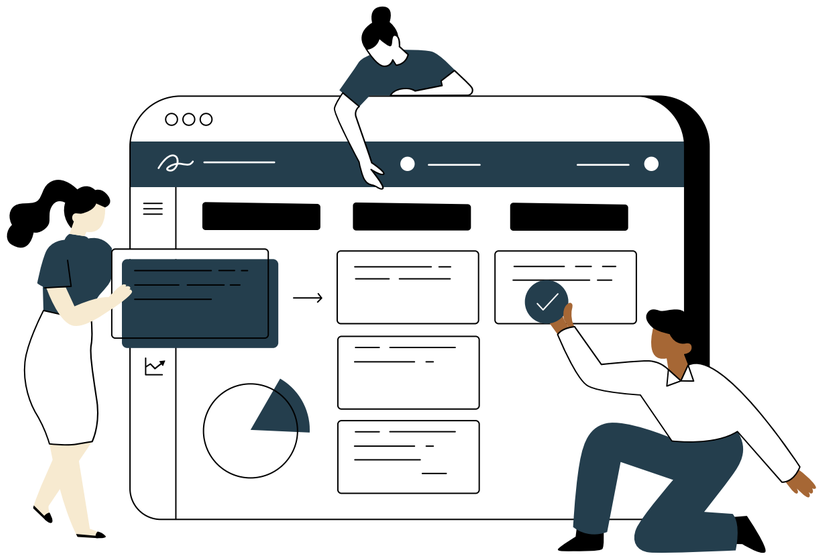
Attrition Count by AgeGroup and OverTime



Average of JobSatisfaction, Average of EnvironmentSatisfaction, Average of WorkLifeBalance and Average of RelationshipSatisfaction by Department



Explanation



1. Donut Chart — Attrition by Job Role

- Displays employee attrition across different job roles
- Helps identify roles with the highest turnover

2. Clustered Bar Chart — Attrition by Age Group & Overtime

- Compares attrition across age groups based on overtime status
- Helps analyze the impact of overtime on employee turnover

3. Clustered Column Chart — Satisfaction Scores by Department

- Compares Job Satisfaction, Environment Satisfaction, Work-Life Balance, and Relationship Satisfaction across departments
- Helps identify departments with lower employee satisfaction levels

SALARY AND PROMOTION

Department

- [ ] Human Resources
- [ ] Research & Development
- [ ] Sales

JobLevel

1 5

6.50K
Average of MonthlyIn...

147
High Risk Employees

7.01
Average of YearsAtCo...

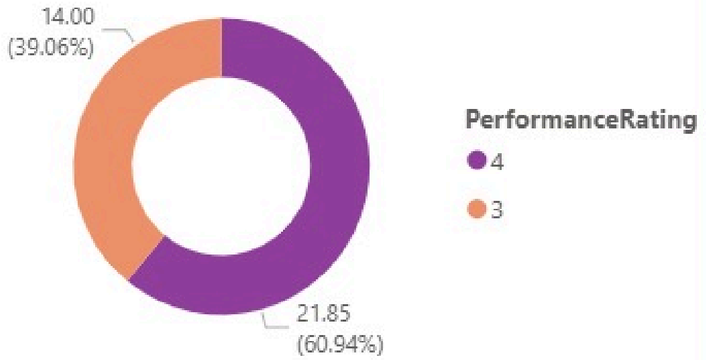
2.19
Average of YearsSinc...

15.21
Average of PercentSalary...

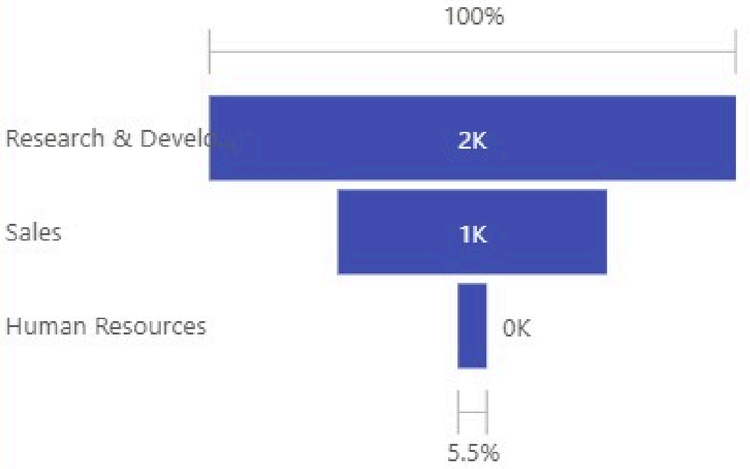
Average of MonthlyIncome by JobLevel



Average of PercentSalaryHike by PerformanceRating



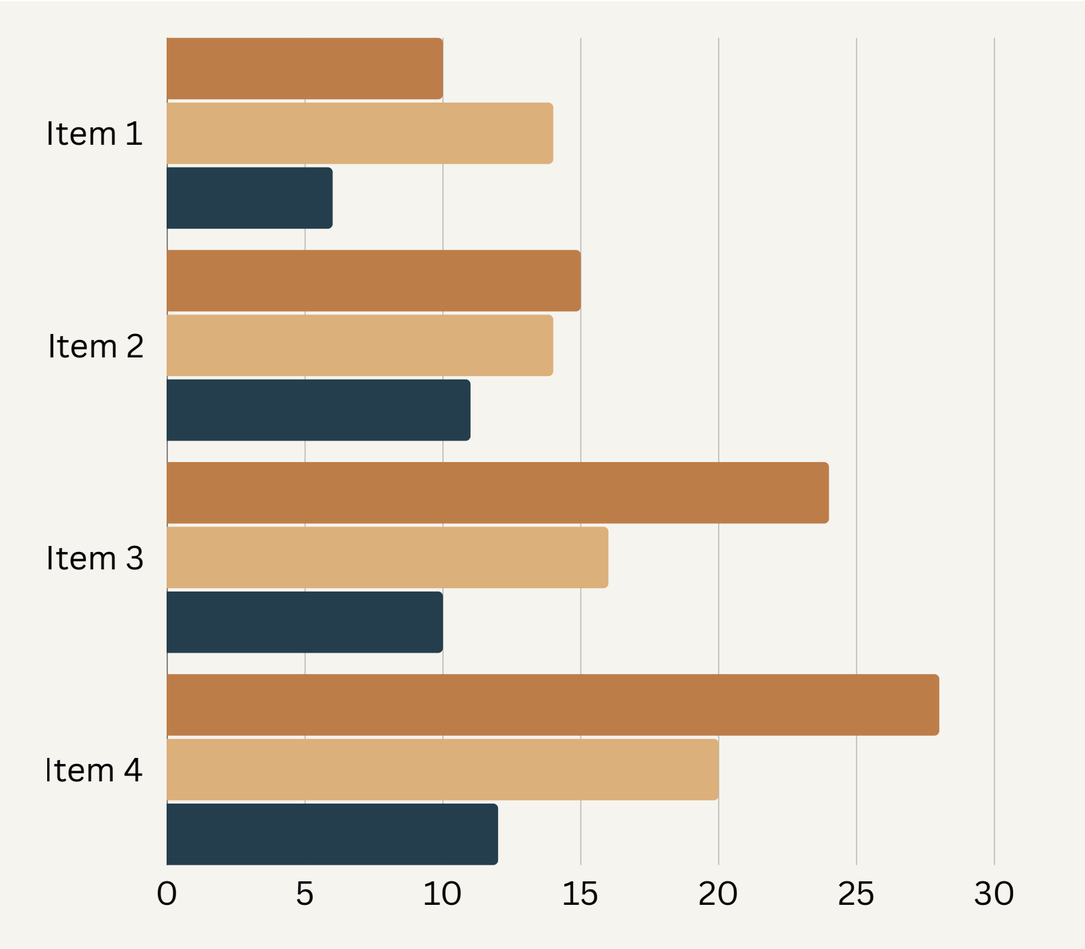
Sum of YearsSinceLastPromotion by Department



Explanation



1. **Line Chart — Average Salary by Job Level**
   - Shows salary distribution across different job levels
   - Helps analyze compensation growth with seniority
2. **Donut Chart — Salary Hike % by Performance Rating**
   - Compares salary hike percentages based on employee performance ratings
   - Helps evaluate reward patterns for performance
3. **Funnel Chart — Years Since Promotion by Department**
   - Displays promotion gaps across departments
   - Helps identify departments with delayed promotions



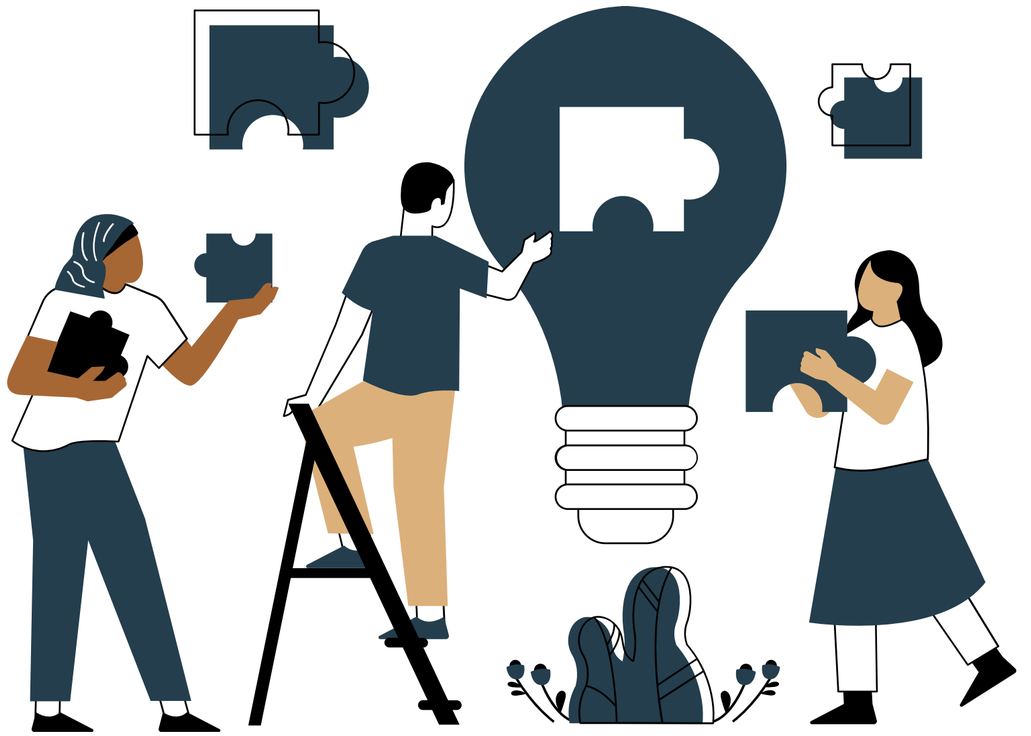
Insights

Attrition & Performance

A Employees aged 18–30 working overtime show the highest attrition levels

B Laboratory Technicians and Sales Executives have the highest employee turnover

C Human Resources shows comparatively lower employee satisfaction scores

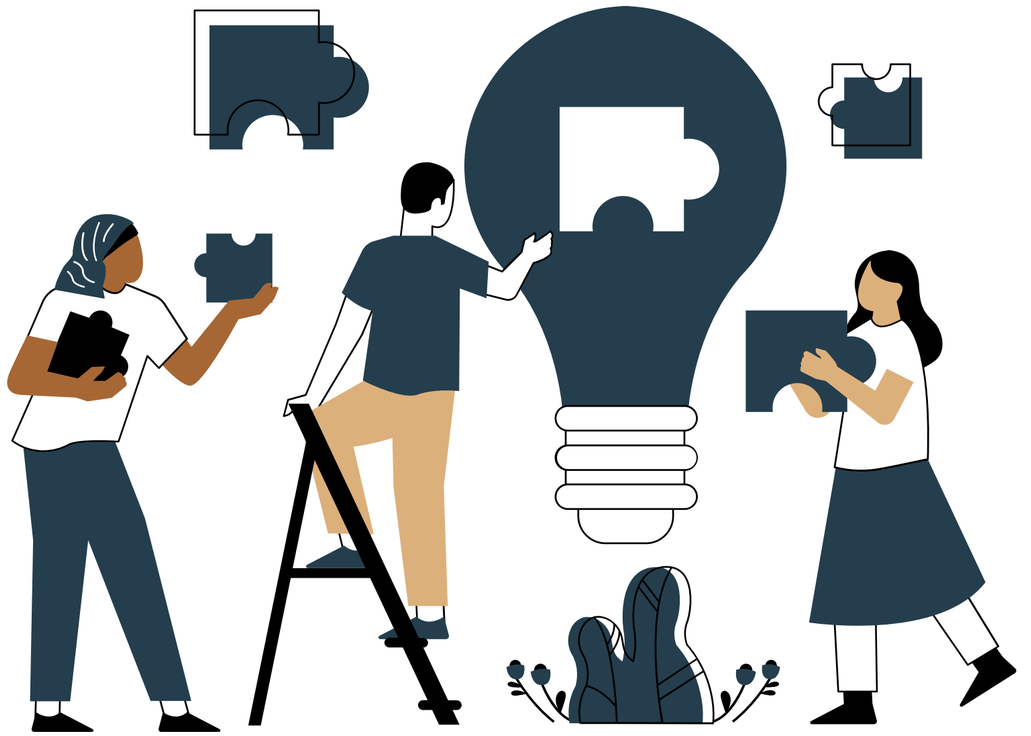


Salary & Promotion

A Average salary increases significantly with higher job levels

B Employees with better performance ratings receive higher salary hikes

C Research & Development shows the highest promotion delays among departments



Thank You